

Project Report: Multi-Agent DevOps Copilot - Agentic AI Engineer

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Family:	Agentic AI Engineer
Level:	Advanced
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Planner + executor + reviewer agents orchestrate deployment tasks via MCP tools with safety constraints.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Advanced]

Chapter focus: Project Overview** *(Level: Advanced)

Multi-Agent DevOps Copilot** is a advanced-level end-to-end project for Agentic AI Engineer. **Duration:** 5-6 weeks. Planner + executor + reviewer agents orchestrate deployment tasks via MCP tools with safety constraints. **Tools:** LangGraph, MCP, OpenAI, Langfuse.

Code Reference:

Objective: 3 specialized agents (plan, execute, review)
Objective: MCP tools for git, CI, and logs
Objective: Safety: block destructive ops without approval
Objective: Full trace visualization

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Agentic AI Engineer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Agentic AI Engineer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: MCP [Advanced]

Chapter focus: Phase: MCP *(Level: Advanced)**

Connect MCP servers. Complete these tasks: Git tools; CI API; Log reader. Deliverable for this phase: MCP tool catalog.

Code Reference:

- [] Git tools
- [] CI API
- [] Log reader

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase MCP is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Agentic AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the MCP tool catalog to the teacher.

Chapter 3: Phase: Agents [Advanced]

Chapter focus: Phase: Agents *(Level: Advanced)**

Multi-agent graph. Complete these tasks: Planner; Executor; Reviewer. Deliverable for this phase: Agent interaction diagram.

Code Reference:

- [] Planner
- [] Executor
- [] Reviewer

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Agents is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Agentic AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Agent interaction diagram to the teacher.

Chapter 4: Phase: Safety [Advanced]

Chapter focus: Phase: Safety** *(Level: Advanced)

Destructive op gate. Complete these tasks: Block list; Approval flow. Deliverable for this phase: Safety test cases passed.

Code Reference:

- [] Block list
- [] Approval flow

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Safety is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Agentic AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Safety test cases passed to the teacher.

Chapter 5: Phase: Trace [Advanced]

Chapter focus: Phase: Trace** *(Level: Advanced)

Full run visualization. Complete these tasks: LangSmith/Langfuse; Step replay. Deliverable for this phase: Trace walkthrough video.

Code Reference:

- [] LangSmith/Langfuse

- [] Step replay

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Trace is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Agentic AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Trace walkthrough video to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Advanced]

Chapter focus: Final Deliverables and Review** *(Level: Advanced)

Submit all final deliverables: Multi-agent system; MCP config; Safety tests; Trace demo video. Skills demonstrated: Multi-agent, MCP, Safety. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

Deliverable: Multi-agent system
Deliverable: MCP config
Deliverable: Safety tests
Deliverable: Trace demo video

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Multi-agent system and passes the rubric.